PHYS4055/7045

Photonic Quantum Information

Jacquiline Romero

Plan…

What is a photon?

Photonic encodings

Dual-rail encoding

Gates and how to implement them

Overview of photonic quantum computing

Why photons?

The world is cold at optical frequencies…

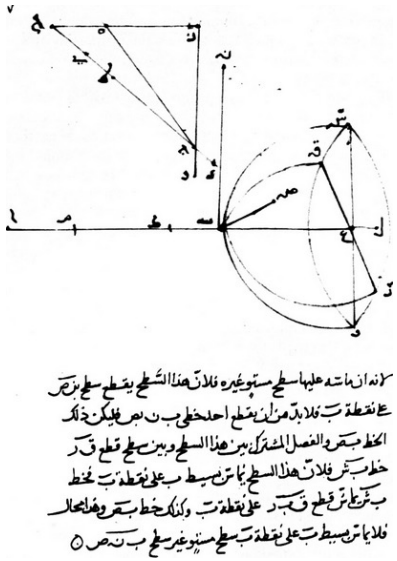
The frequency of photonic excitations is $\omega \sim 10^{14}$

At room temperature: $\frac{\hbar\omega}{k_B T} \sim 10$

The average number of background thermal photons is: $\bar{n}(\omega) = \frac{1}{e^{\frac{\hbar\omega}{k_B T}} - 1} \sim 0$

Quantum effects manifest at room temperature.

Some history



Alhazen
9th century

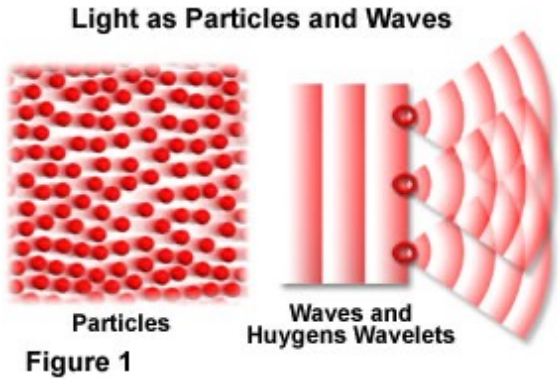
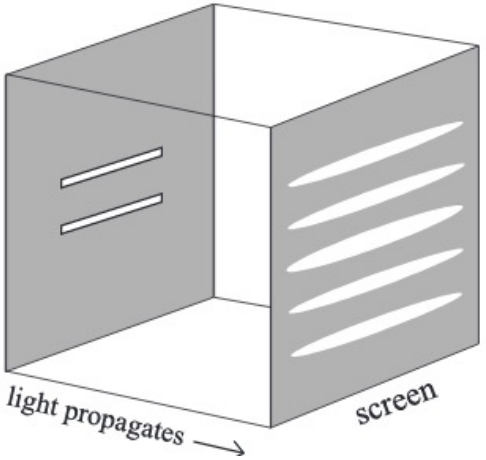


Figure 1

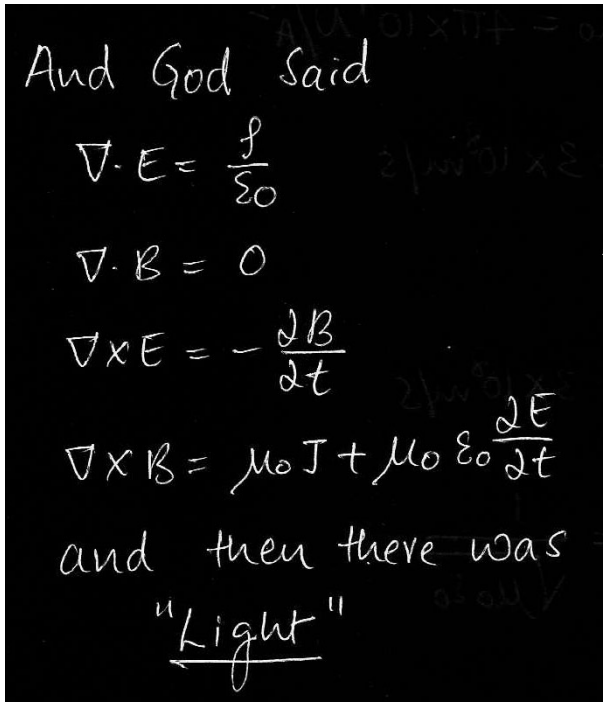
Newton vs Huygens
17th century



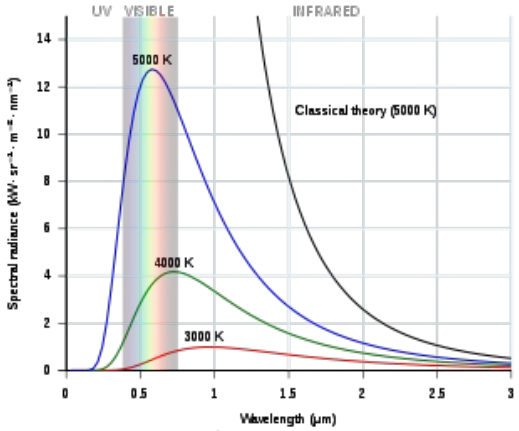
Young's double slit experiment
1800

wikipedia.org; micro.magnet.fsu.edu; OPN Trends 2003

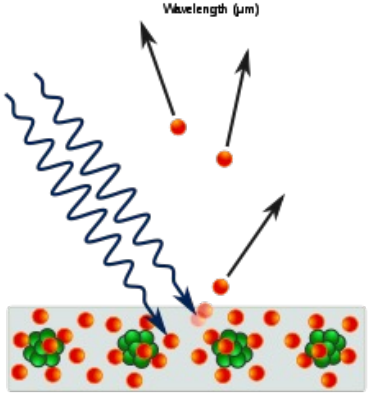
Some history



Maxwell's equations
1873



Planck's law
1900

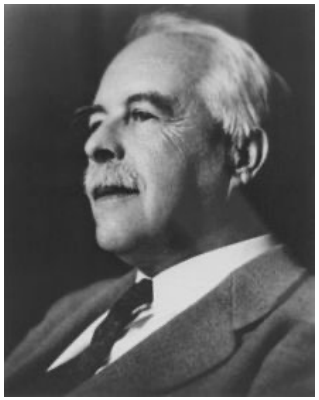


Photoelectric effect – Einstein 1905

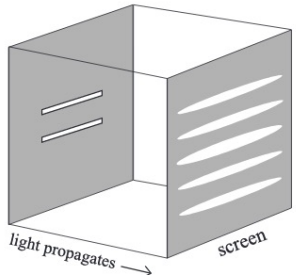
discreteness in energy more than a "localisation"

readingfeynman.org, wikipedia.org

"Photon" was controversial



Bohr, de Broglie, Schrödinger, Dirac, Heisenberg,
Quantum mechanics
1920s



Semiclassical theory can describe the photoelectric effect (!)

PHOTON coined by Gilbert Lewis
1926

…thought of it as
a "hypothetical new atom"



Willis Lamb, Nobel 1955

Appl. Phys. B 60, 77–84 (1995)

Applied Physics B Lasers and Optics
© Springer-Verlag 1995

Anti-photon

W.E. Lamb, Jr.

Optical Sciences Center, University of Arizona, Tucson, AZ 85721, USA

Received: 23 July 1994 / Accepted: 18 September 1994

wikipedia.org



Appl. Phys. B 60, 77–84 (1995)

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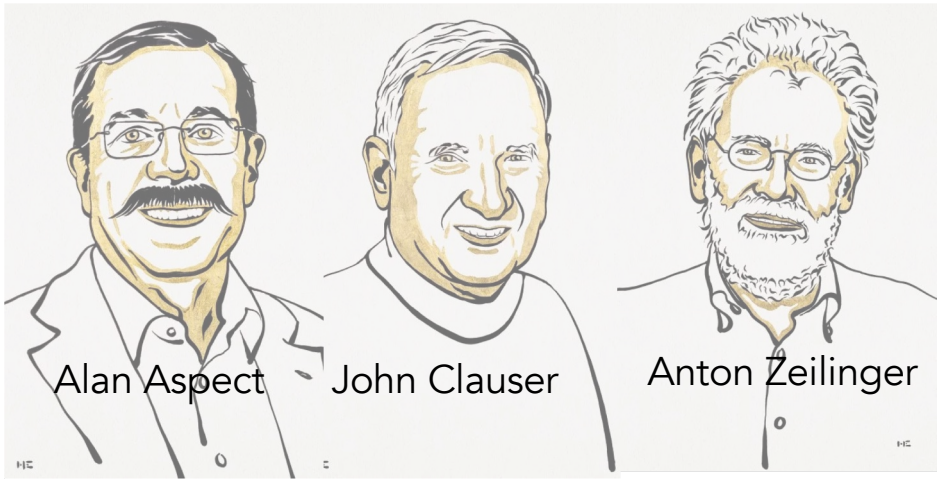
Received: 23 July 1994 / Accepted: 18 September 1994

At the first of the 1960's Rochester Coherence Conferences, I suggested that a license be required for use of the word "photon", and offered to give such a license to properly qualified people. My records show that nobody working in Rochester, and very few other people elsewhere, ever took out a license to use the word "photon". It reminds me that there was once a phlogiston theory of heat, which began to go out of style about the time that people at Cambridge University stopped using the corpuscular optics of Newton.









for experiments with entangled photons,
establishing the violation of Bell inequalities and pioneering quantum information science

What is a photon?

A photon is a single particle excitation of an optical field mode, where the mode operators satisfy the boson commutation relation:

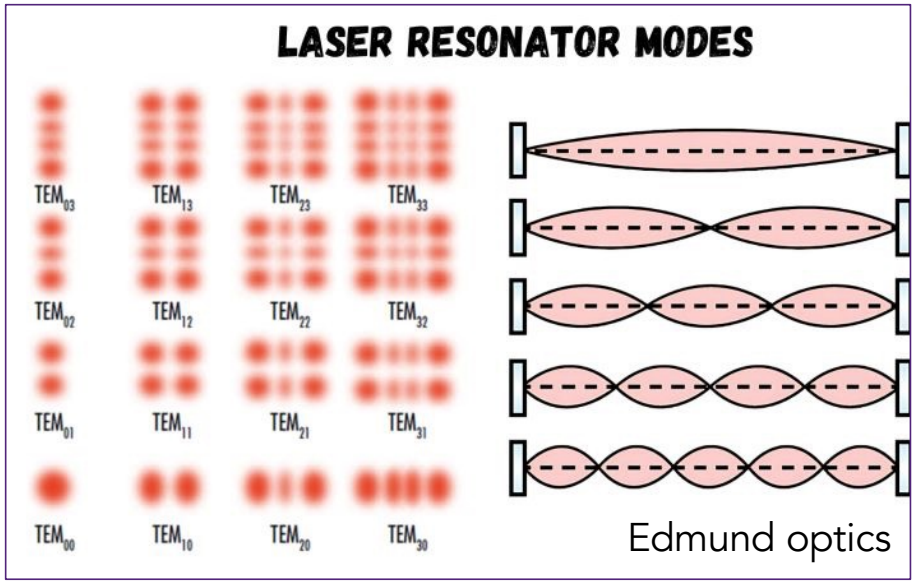
$$\left[a_k, a_{k'}^\dagger\right] = \delta_{kk'}$$

$$|1\rangle = \sum_k f_k a_k^\dagger |0\rangle \quad f_k \in \mathbb{C}$$

The Hilbert space of the photon is associated with the orthonormal modes of the electromagnetic field.

The total Hilbert space of the electromagnetic field is a tensor product Hilbert space of many modes.

Where is the photon?



The photon is common to the cavity, because the mode is common to the cavity.

What if there is no cavity?

What is a photon? (*Experimentalist's answer)*

A photon is what a photodetector detects – Roy Glauber



Perkin Elmer (Excelitas)

Discrete variable vs continuous variable

The operator with a discrete spectrum is the number operator for mode k,

$$a_k^\dagger a_k |n\rangle_k = n_k |n\rangle_k \quad n = 0, 1, 2, \ldots$$

The operator with a continuous spectrum are the quadrature operators for mode k,

$$q_k = \frac{1}{\sqrt{2}}\left(a_k + a_k^\dagger\right), \quad p_k = -i\frac{1}{\sqrt{2}}\left(a_k - a_k^\dagger\right) \quad [q_k, p_{k'}] = i\delta_{kk'}$$

Discrete variable encoding

single-rail

$$|\psi\rangle = \sum_j c_j |j\rangle$$

Dual-rail (d=2)

$|0\rangle$ no photon

$|1\rangle$ 1 photon

$$|\mathbf{0}\rangle_L = a_0^\dagger |0,\ldots,0,0\rangle_{012\ldots d-1} = |0,\ldots,0,1\rangle_{012\ldots d-1} = |0\rangle$$
$$|\mathbf{1}\rangle_L = a_1^\dagger |0,\ldots,0,0\rangle_{012\ldots d-1} = |0,\ldots,1,0\rangle_{012\ldots d-1} = |1\rangle$$

$$|0\rangle = a_1^\dagger |0,0\rangle_{1,2} = |1,0\rangle_{1,2}$$
$$|1\rangle = a_2^\dagger |0,0\rangle_{1,2} = |0,1\rangle_{1,2}$$

d-rail encoding $|\mathbf{d-1}\rangle_L = a_{d-1}^\dagger |0,\ldots,0,0\rangle_{012\ldots d-1} = |1,\ldots,0,0\rangle_{012\ldots d-1} = |d-1\rangle.$

Continuous variable encoding

$$|\psi\rangle = \int_{-\infty}^{+\infty} dq\psi(q)|q\rangle = \int_{-\infty}^{+\infty} dp\tilde{\psi}(p)|p\rangle$$

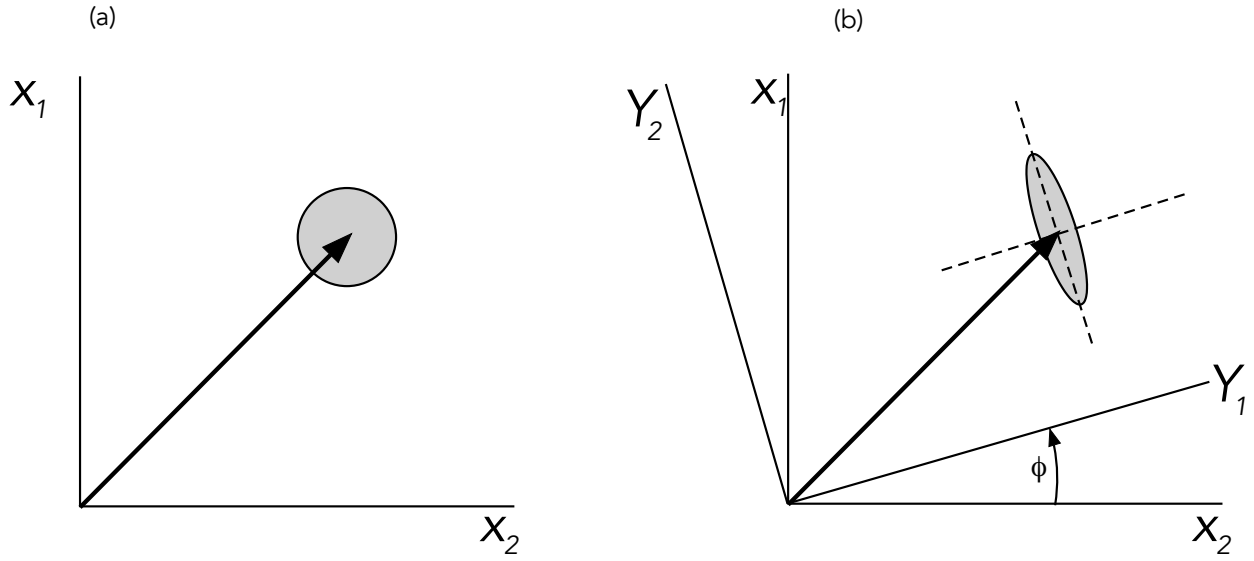
where

$$\tilde{\psi}(p) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dq e^{-ipq}\psi(q)$$

$$\langle q' \mid q\rangle = \delta\left(q' - q\right), \langle p' \mid p\rangle = \delta\left(p' - p\right)$$

Note that q and p are not physical states. (Why?)

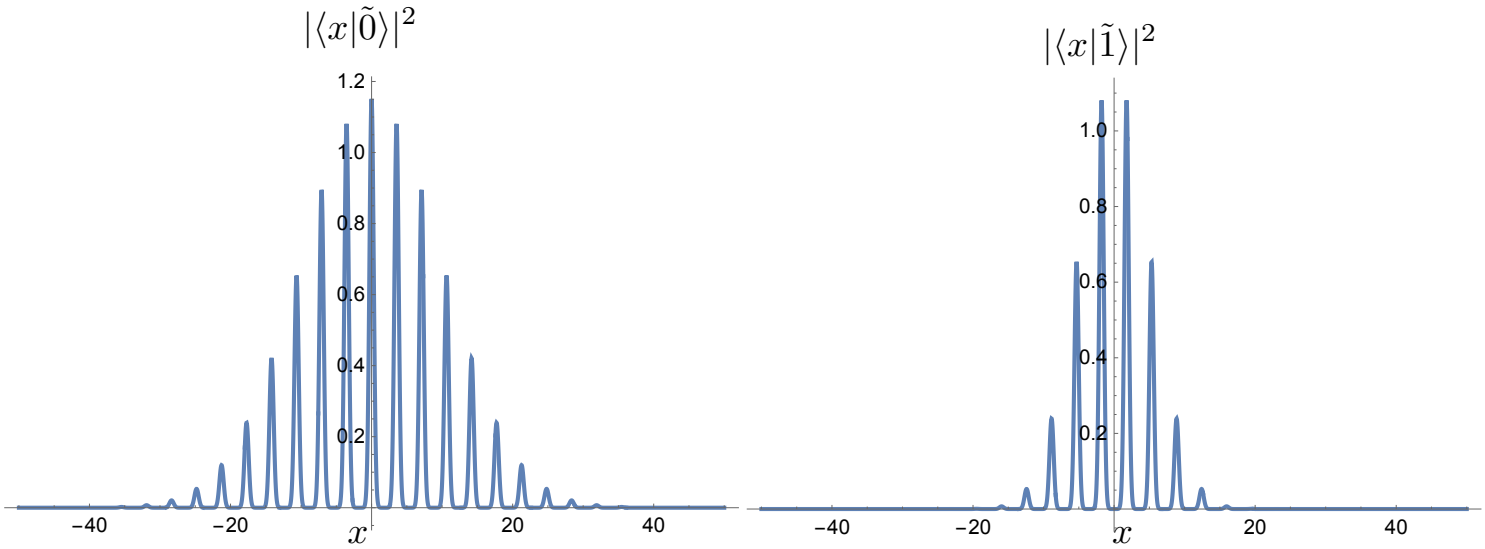
Continuous variable encoding



coherent

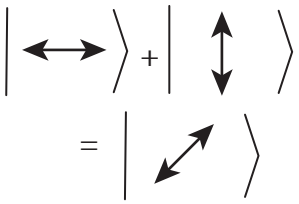
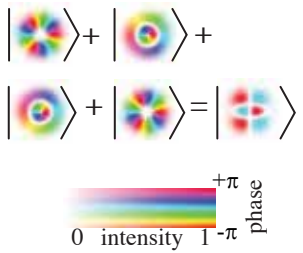
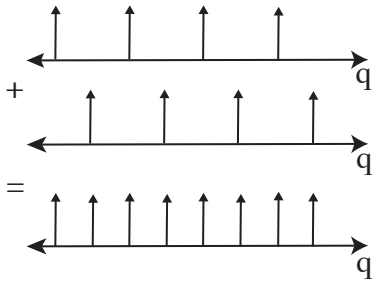
squeezed

Gottesman-Kitaev-Preskill encoding

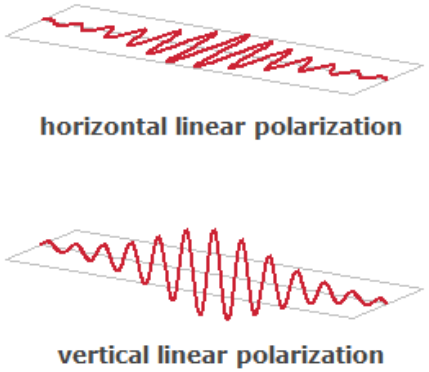


Gottesman, Kitaev, Preskill, PRA 2001

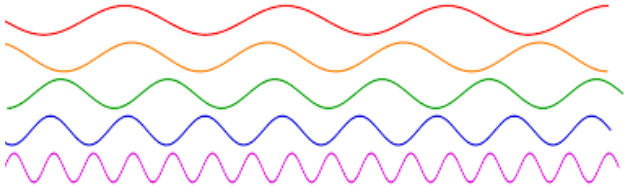
Discrete variable vs continuous variable

	Qubit (DV)	Qudit (DV)	Continuous Variable (CV)
basis vectors	$\{\|0\rangle, \|1\rangle\}$	$\{\|0\rangle, \|1\rangle, ..., \|d-1\rangle\}$	$\{\|q\rangle\} \in \mathbb{R}$
orthogonality	$\langle \ell \| m \rangle = \delta_{\ell,m}$ $\ell, m \in \{0, 1\}$	$\langle \ell \| m \rangle = \delta_{\ell,m}$ $\ell, m \in \{0, 1, \ldots, d-1\}$	$\langle q_1 \| q_2 \rangle = \delta(q_1 - q_2)$ $q_1, q_2 \in \mathbb{R}$
quantum state	$\|\psi\rangle = c_0\|0\rangle + c_1\|1\rangle$	$\|\psi\rangle = c_0\|0\rangle + c_1\|1\rangle + \ldots c_{d-1}\|d-1\rangle$	$\|\psi\rangle = \int_{-\infty}^{+\infty} dq \psi(q) \|q\rangle$
physical example	polarisation 	transverse spatial modes 	quadrature 

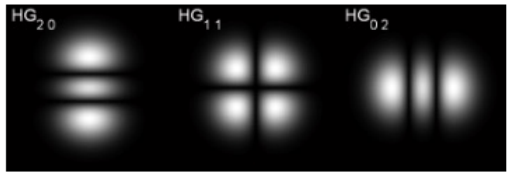
Photonic properties for qubit (qudit) states



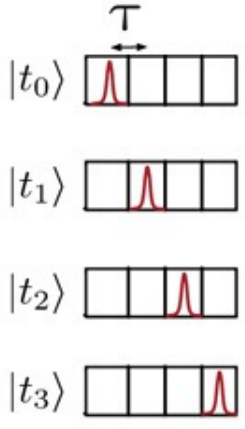
polarisation



wavelength/frequency

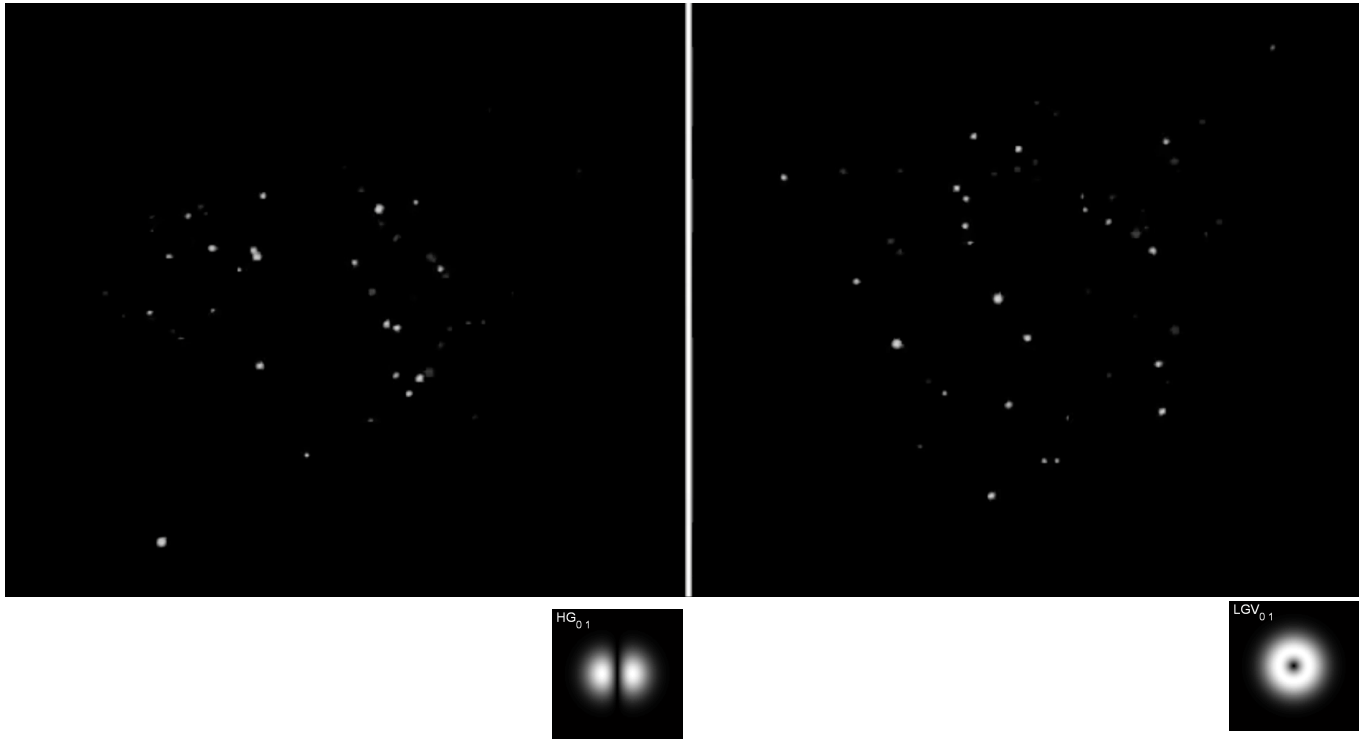


transverse mode/shape
(OAM/twisted/structured light)



time bins

The shape is the probability of detecting a photon at (x,y)



Fickler et al, Sci. Rep. 3, 1914 (2013)

Quantum states from the shape of light

$$|\psi\rangle = \alpha\,|\,\rangle + \beta\,|\,\rangle$$



$$= \alpha|\,\rangle + \beta|\,\rangle$$

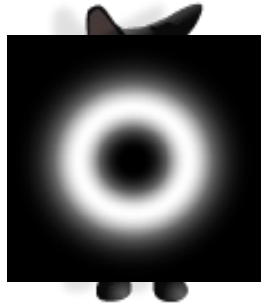
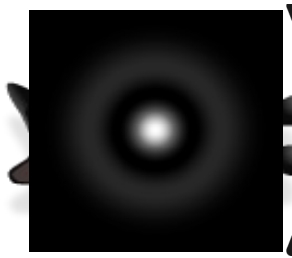


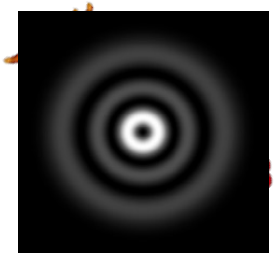


qubits

Quantum states from the shape of light are not limited to qubits

ZOO

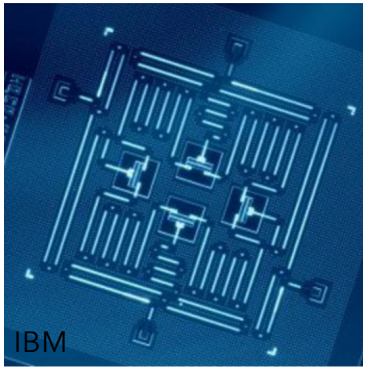
$$= \alpha \left| \text{  } \right\rangle + \beta \left| \text{  } \right\rangle$$

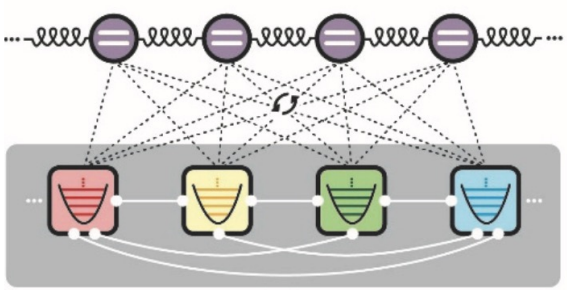
$$+ \gamma \left| \text{  } \right\rangle + \eta \left| \text{  } \right\rangle$$

Qudits@UQ

Aside: Matter-based vs photonic quantum computer

matter

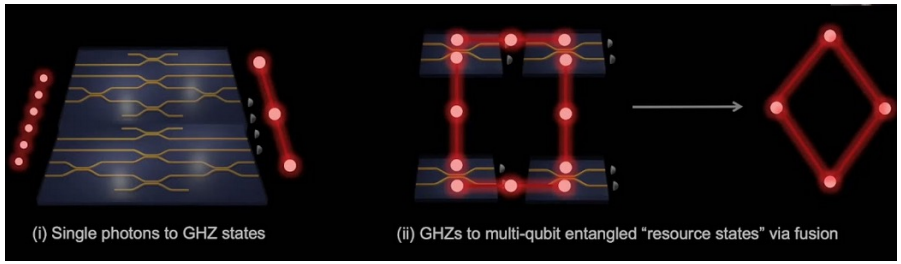




- Examples: spin, superconducting, ions, neutral atoms
- Qubits are fixed in place
- Time-dependent controls are switched on and off in a particular order
- Closer to conventional silicon computer

photonic

PsiQuantum



- Example: photons, electromagnetic field
- Pulses of light moving through fibres or waveguides
- Entangled qubits are continuously created and measured.
- Closer to an all-optical network.

Polarisation qubits

$$\hat{a}_F^\dagger \to \hat{a}_F^\dagger \quad \text{and} \quad \hat{a}_S^\dagger \to -\hat{a}_S^\dagger. \tag{5.15}$$

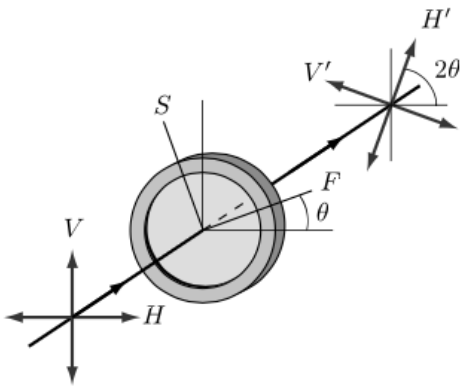


Fig. 5.1. The effect of a 'half-wave plate' (HWP) on a linearly polarized beam of light. Axes parallel and perpendicular to the optical axis of the half-wave plate are indicated by *F* and *S*, respectively. A spatial rotation over an angle θ of the wave plate induces a polarization rotation over an angle 2θ.

Kok and Lovett, 2010

Polarisation qubits

$$U_{\mathrm{HWP}}(\theta) = U_Y(2\theta) = \exp(2i\theta Y) = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$$

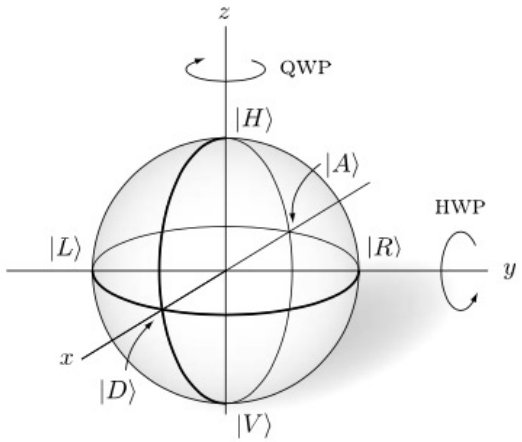


Fig. 5.2. The Bloch sphere for a single-photon polarization qubit. The effects of a 'half-wave plate' (HWP) and a 'quarter-wave plate' (QWP) are indicated as rotations around the *y* and the *z* axis, respectively.

Kok and Lovett, 2010

Polarisation qubits

$$U_{\mathrm{QWP}} = \exp\left(-\frac{i\pi}{4}Z\right) = \exp(-i\pi/4)\begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix}$$

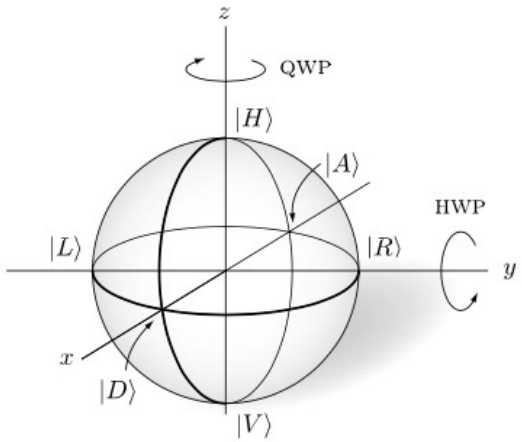


Fig. 5.2. The Bloch sphere for a single-photon polarization qubit. The effects of a 'half-wave plate' (HWP) and a 'quarter-wave plate' (QWP) are indicated as rotations around the *y* and the *z* axis, respectively.

TABLE 1. Jones Matrices of Common Optical Devices

Vertical Linear Polarizer	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	Right Circular Polarizer	$\frac{1}{2}\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$
Horizontal Linear Polarizer	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	Left Circular Polarizer	$\frac{1}{2}\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$
Linear Polarizer at 45°	$\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	Quarter-wave plate, fast axis vertical	$\begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$ Retardation λ/4
Lossless fiber transmission	$\begin{bmatrix} e^{i\phi}\cos\theta & -e^{-i\psi}\sin\theta \\ e^{i\psi}\sin\theta & e^{-i\phi}\cos\theta \end{bmatrix}$	Quarter-wave plate, fast axis horizontal	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$ Retardation λ/4

Half-wave plate, fast axis horizontal $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ Retardation λ/2

Half-wave plate, fast axis vertical $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ Retardation λ/2

General retarder, fast axis horizontal $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}$ In terms of waves (wavelength λ), this is a retarder $\lambda^{*}\varphi/2\pi$

Arbitraty rotation can be done with sequence of HWP and QWP:

$$U_{\mathrm{QWP}}(\theta) = e^{-i\pi/4}\left[\cos\left(\frac{\pi}{4}\right)\mathbf{I} + i\sin\left(\frac{\pi}{4}\right)\left(\cos(2\theta)\sigma_z + \sin(2\theta)\sigma_x\right)\right]$$

$$U_{\mathrm{HWP}}(\theta) = -i\left(\cos(2\theta)\sigma_z + \sin(2\theta)\sigma_x\right)$$

Fast axis oriented at an angle **θ** wrt to horizontal axis

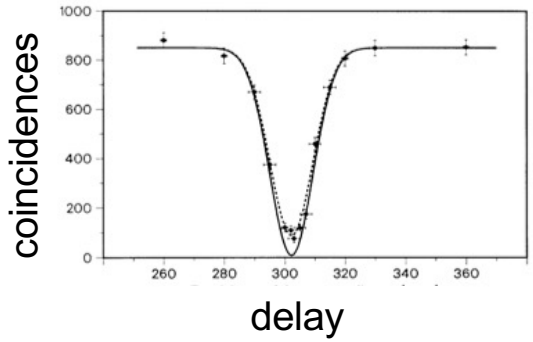
Exercise: generalized two-mode transformations

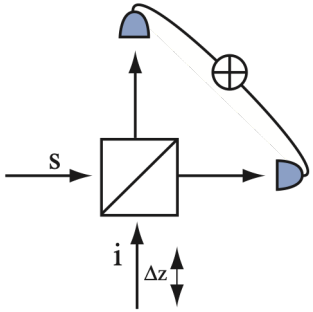
Exercise: generalized two-mode transformations

VOLUME 59, NUMBER 18 PHYSICAL REVIEW LETTERS 2 NOVEMBER 1987

Measurement of Subpicosecond Time Intervals between Two Photons by Interference

C. K. Hong, Z. Y. Ou, and L. Mandel
Department of Physics and Astronomy, University of Rochester, Rochester, New York 14627
(Received 10 July 1987)





Coincidence measurement - Is there a click on BOTH photon detectors?

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A

MATHEMATICAL, PHYSICAL AND ENGINEERING SCIENCES

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Opinion piece

The not quite Loudon–Fearn–Rarity–Tapster dip and its impact on the development of photonic quantum information

John G. Rarity

Published: 24 December 2024 https://doi.org/10.1098/rsta.2024.0393

2. Some mathematical details

The discovery of the destructive interference at a beamsplitter showed the utility of the creation and annihilation operator [23] formalism for describing the interactions and interference of quantum states in classical apparatus. For the purposes of this paper, we introduce the creation operators $\boldsymbol{a}_i^+$ for mode i which, when operating on a vacuum state, populates mode i with a single photon

$$\boldsymbol{a}_i^+|0\rangle = |1\rangle_i \tag{2.1}$$

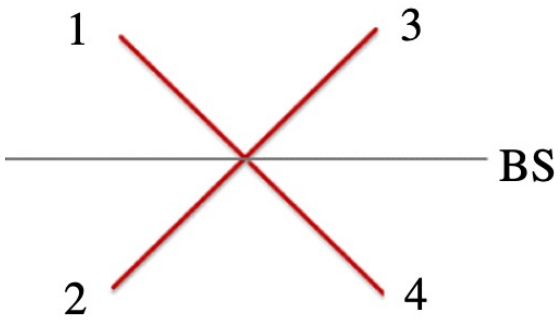
Repeated operations with suitable normalization increase the photon number as in

$$\frac{\boldsymbol{a}_i^{+N}}{\sqrt{N!}}|\,0\rangle = |\,N\rangle_i\,. \tag{2.2}$$

In the opposite direction, annihilation operators $\boldsymbol{a}_i$ reduce the photon number by one. The creation and annihilation operators have commutation relation

$$\left[\boldsymbol{a}_i, \boldsymbol{a}_j^+\right] = \delta(i, j)\,. \tag{2.3}$$

Rarity, Phil. Trans. R. Soc. A 2024



$$|\Psi_{in}\rangle = |1\rangle_1|1\rangle_2 = \boldsymbol{a_1^+ a_2^+}|0\rangle .$$

$$\boldsymbol{a_1^+ \rightarrow t a_4^+ + i r a_3^+ : a_2^+ \rightarrow t a_3^+ + i r a_4^+ .}$$

$$|\Psi_{out}\rangle = \left(t\boldsymbol{a_4^+} + \boldsymbol{i r a_3^+}\right)\left(t\boldsymbol{a_3^+} + \boldsymbol{i r a_4^+}\right)|0\rangle = \boxed{\left(t^2 - r^2\right)|1\rangle_3|1\rangle_4} + \sqrt{2} i r t\,|2\rangle_3 + \sqrt{2} i r t\,|2\rangle_4 .$$

0 if t=r

VOLUME 73, NUMBER 1 PHYSICAL REVIEW LETTERS 4 JULY 1994

Experimental Realization of Any Discrete Unitary Operator

Michael Reck and Anton Zeilinger
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(Received 11 February 1994)

An algorithmic proof that any discrete finite-dimensional unitary operator can be constructed in the laboratory using optical devices is given. Our recursive algorithm factorizes any $N \times N$ unitary matrix into a sequence of two-dimensional beam splitter transformations. The experiment is built from the corresponding devices. This also permits the measurement of the observable corresponding to any discrete Hermitian matrix. Thus optical experiments with any type of radiation (photons, atoms, etc.) exploring higher-dimensional discrete quantum systems become feasible.

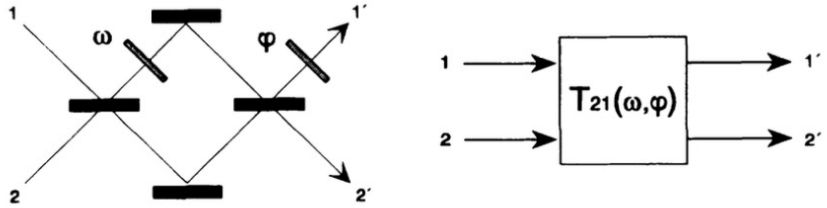


FIG. 1. A Mach-Zehnder interferometer can be used instead of a variable reflectivity beam splitter as the basic building block of any $N \times N$ unitary matrix. On the left is one experimental realization of the device using two 50:50 beam splitters, two mirrors, and two phase shifters. The Mach-Zehnder interferometer can be represented by the abstract four-port device on the right. Two parameters (ϕ, ω) of the transformation T are set in the device.

nature

Article | Published: 04 January 2001

A scheme for efficient quantum computation with linear optics

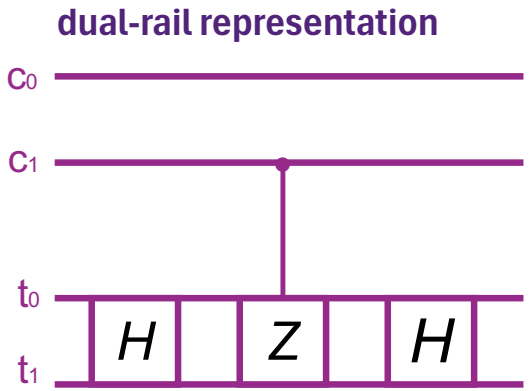
E. Knill ✉, R. Laflamme & G. J. Milburn

Nature **409**, 46–52 (2001) | Cite this article

Nonlinear two-qubit gate

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$\boldsymbol{CNOT}$ or $\boldsymbol{CX} = \boldsymbol{H}.\boldsymbol{CZ}.\boldsymbol{H}$, for d=2



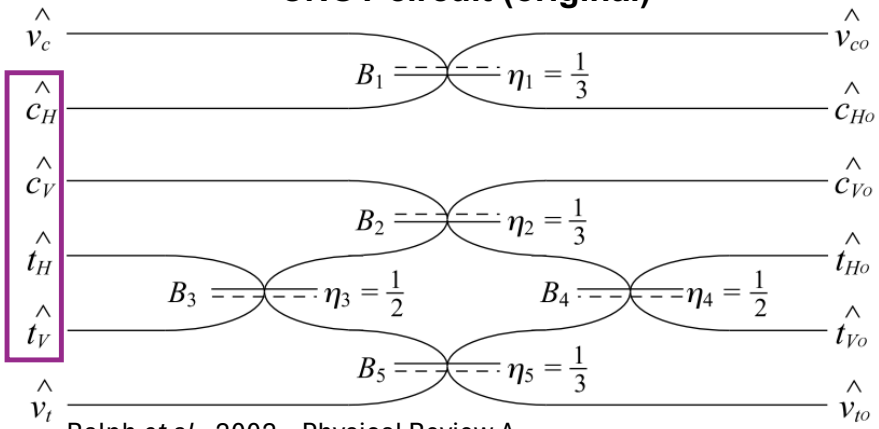
Basis State	Output
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|01\rangle$
$\|10\rangle$	$\|11\rangle$
$\|11\rangle$	$\|10\rangle$

Slides from Y. Allo

Multi-qubit operation, e.g., CNOT gate

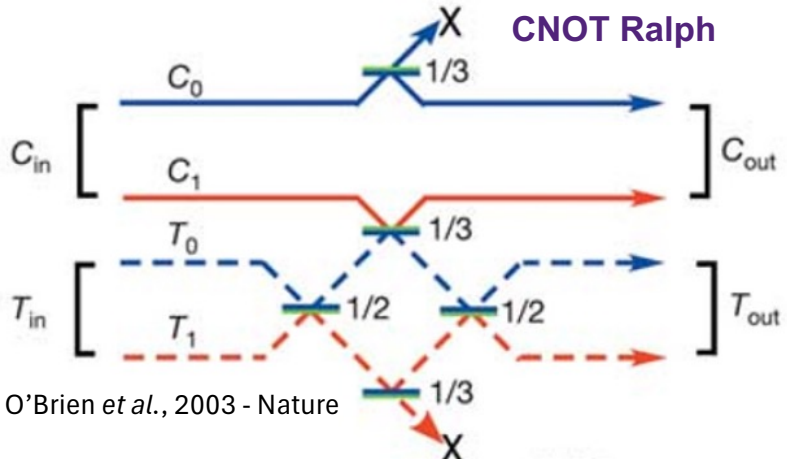
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CNOT circuit (original)



Ralph *et al.*, 2002 – Physical Review A

CNOT Ralph

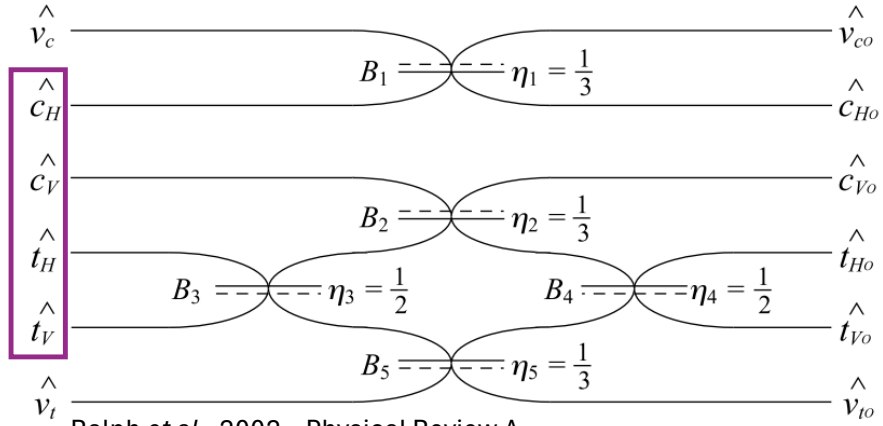


O'Brien *et al.*, 2003 - Nature

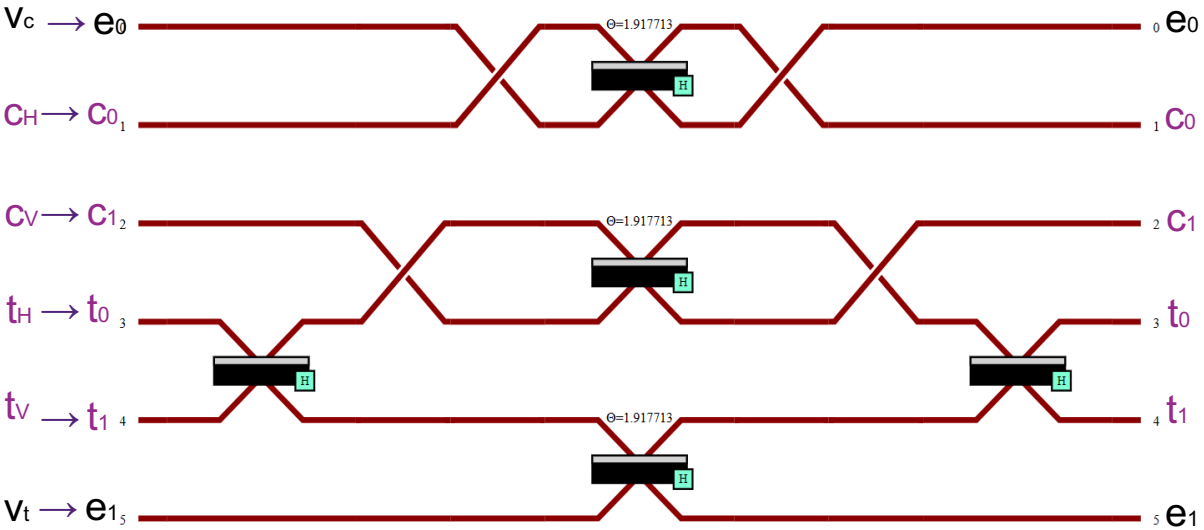
2-qubit operation, e.g., CNOT gate

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CNOT circuit



Ralph *et al.*, 2002 – Physical Review A




```
Input |00⟩
  |1,0,1,0,0,0>: 22.45%
  |1,0,0,1,0,0>: 22.11%
  |1,0,0,0,0,1>: 22.45%
  |0,1,1,0,0,0>: 11.06%
  |0,1,0,1,0,0>: 10.89%
  |0,1,0,0,0,1>: 11.06%
Input |01⟩
  |1,0,1,0,0,0>: 22.45%
  |1,0,0,1,0,0>: 22.11%
  |1,0,0,0,0,1>: 22.45%
  |0,1,1,0,0,0>: 11.06%
  |0,1,0,0,1,0>: 10.89%
  |0,1,0,0,0,1>: 11.06%
Input |10⟩
  |0,0,2,0,0,0>: 22.11%
  |0,0,1,1,0,0>: 0.0%
  |0,0,1,0,1,0>: 11.22%
  |0,0,1,0,0,1>: 11.06%
  |0,0,0,2,0,0>: 22.11%
  |0,0,0,1,1,0>: 11.06%
  |0,0,0,1,0,1>: 11.22%
  |0,0,0,0,1,1>: 11.22%
Input |11⟩
  |0,0,2,0,0,0>: 22.11%
  |0,0,1,1,0,0>: 11.22%
  |0,0,1,0,1,0>: 0.0%
  |0,0,1,0,0,1>: 11.06%
  |0,0,0,1,1,0>: 11.06%
  |0,0,0,1,0,1>: 11.22%
  |0,0,0,0,2,0>: 22.11%
  |0,0,0,0,1,1>: 11.22%
```

Output

$|e_0, c_0, c_1, t_0, t_1, e_1\rangle$

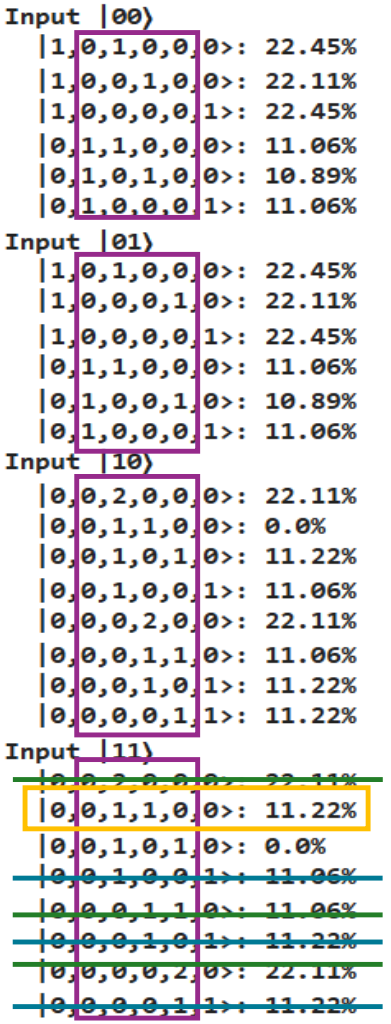
Post-selection

Modelling via Perceval (Quandela)

Post-selection for CNOT qubit

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Basis State	Output	
|00⟩	|00⟩	[1,0,1,0]
|01⟩	|01⟩	[1,0,0,1]
|10⟩	|11⟩	[0,1,0,1]
|11⟩	|10⟩	[0,1,1,0]



Output

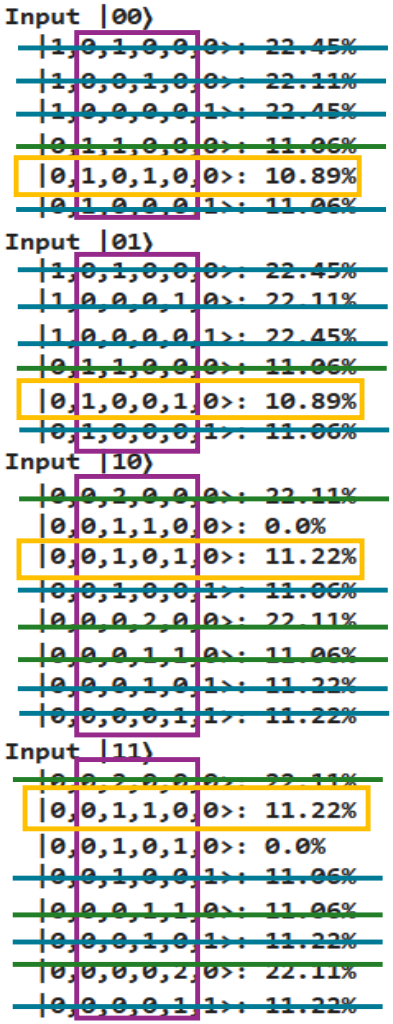
$|e_0, c_0, c_1, t_0, t_1, e_1\rangle$

Post-selection

1. no photon in $e_{0,1}$
2. 1 photon in c,t
3. follow truth table

Post-selection for CNOT qubit

Basis State	Output	
$\|00\rangle$	$\|00\rangle$	[1,0,1,0]
$\|01\rangle$	$\|01\rangle$	[1,0,0,1]
$\|10\rangle$	$\|11\rangle$	[0,1,0,1]
$\|11\rangle$	$\|10\rangle$	[0,1,1,0]



Output

$|e_0, c_0, c_1, t_0, t_1, e_1\rangle$

Post-selection

1. no photon in $e_{0,1}$
2. 1 photon in c,t
3. follow truth table

Post-selection for CNOT qubit ⟶ Probabilistic Result

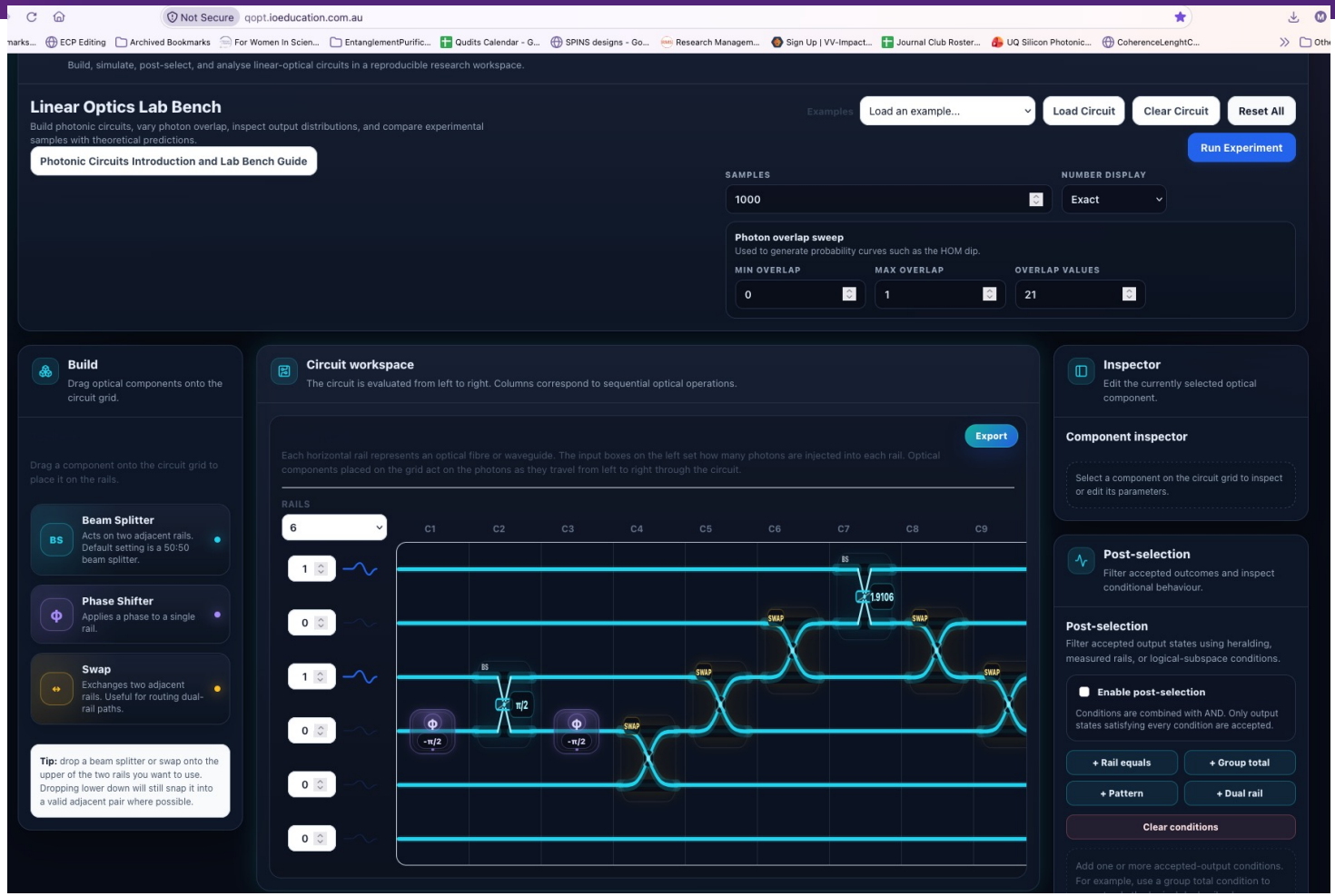
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Basis State	Output	
$\|00\rangle$	$\|00\rangle$	[1,0,1,0]
$\|01\rangle$	$\|01\rangle$	[1,0,0,1]
$\|10\rangle$	$\|11\rangle$	[0,1,0,1]
$\|11\rangle$	$\|10\rangle$	[0,1,1,0]

$$P_{suc} = 11\ \% = 1/9$$

Because of low probabilities–we need many components.This is another motivation to go on-chip.

```
Input |00⟩
  |1,0,1,0,0,0>: 22.45%
  |1,0,0,1,0,0>: 22.11%
  |1,0,0,0,0,1>: 22.45%
  |0,1,1,0,0,0>: 11.06%
  |0,1,0,1,0,0>: 10.89%
  |0,1,0,0,0,1>: 11.06%
Input |01⟩
  |1,0,1,0,0,0>: 22.45%
  |1,0,0,0,1,0>: 22.11%
  |1,0,0,0,0,1>: 22.45%
  |0,1,1,0,0,0>: 11.06%
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  |0,0,1,0,0,1>: 11.06%
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Input |11⟩
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  |0,0,0,1,1,0>: 11.06%
  |0,0,0,1,0,1>: 11.22%
  |0,0,0,0,2,0>: 22.11%
  |0,0,0,0,1,1>: 11.22%
```



qopt.ioeducation.com.au
Patrick Johnson

Quantum information processing

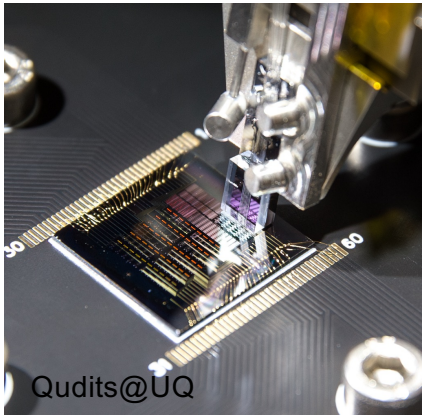
	Generation (Sources)	Processing (Gates)	Measurement and Detection
Free-space	spontaneous parametric down-conversion (SPDC)	Multi-plane light conversion (MPLC) Polarisers, beamsplitters etc.	Mode filtering + single-photon detection
On-chip	spontaneous four-wave mixing (SFWM), SPDC	Interferometers (beamsplitters, phase shifters, gates)	Mode filtering + single-photon detection

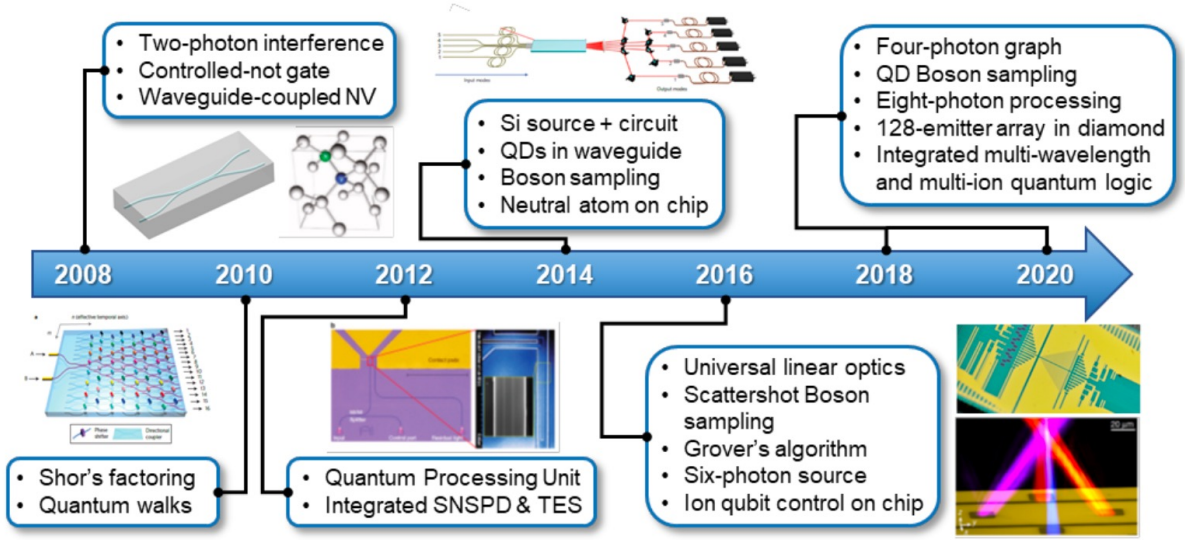
Going on chip…

Improves scalability and efficiency.

Improves stability (less aligning, and aligning again).

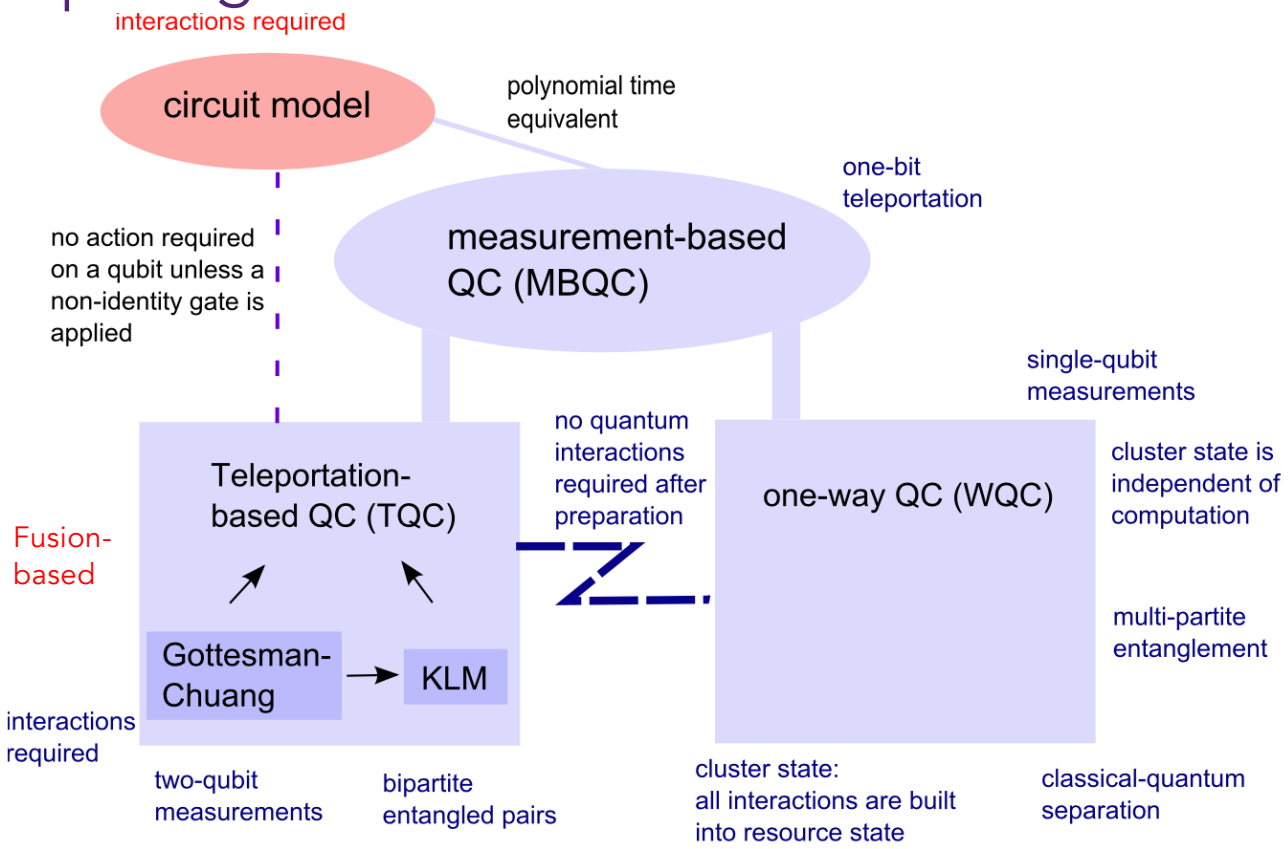
Brings quantum photonics closer to applications.





Moody et al., J Phys Photonics 4, 012501 (2022)

Quantum computing zoo



Barz, J Phys B (2015)

nature communications

The University Of Queensland
AUSTRALIA

Article

https://doi.org/10.1038/s41467-023-36493-1

Fusion-based quantum computation

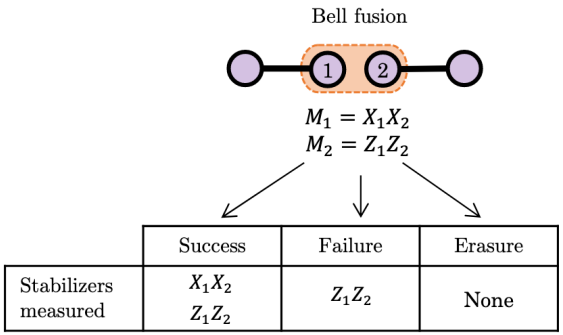
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Accepted: 3 February 2023

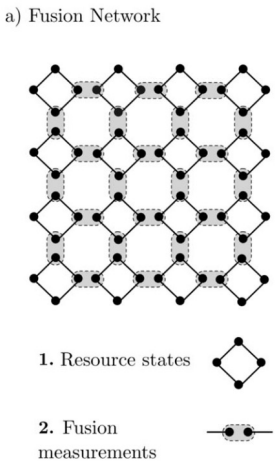
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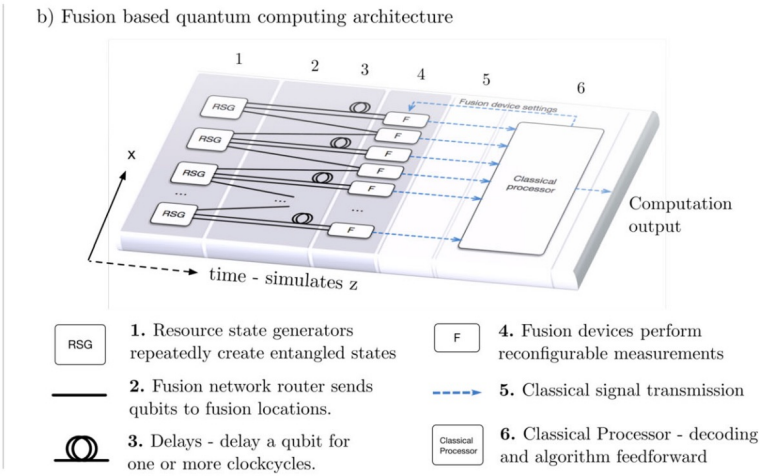
Check for updates

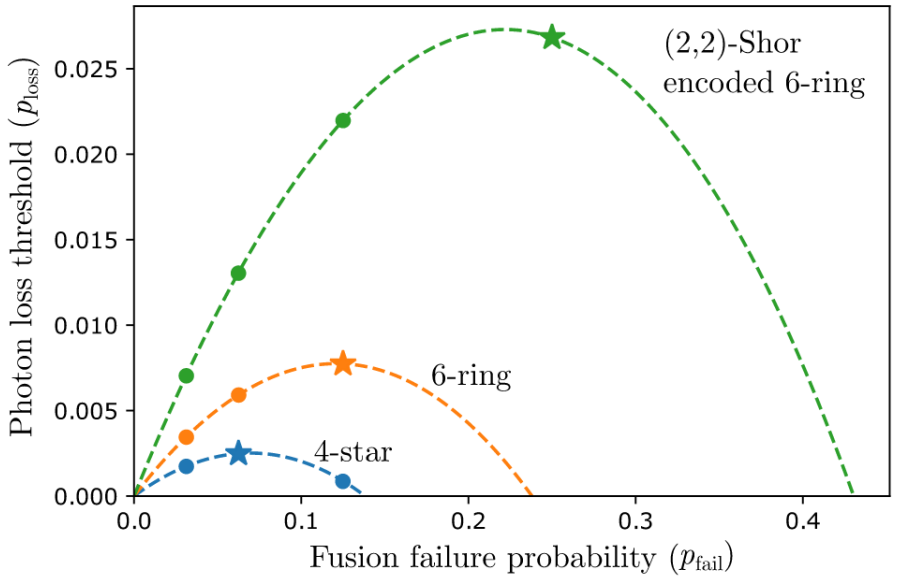
Sara Bartolucci[1], Patrick Birchall[1], Hector Bombín[1,2], Hugo Cable[1], Chris Dawson[1], Mercedes Gimeno-Segovia[1], Eric Johnston[1], Konrad Kieling[1], Naomi Nickerson[1,2] ✉, Mihir Pant[1,2] ✉, Fernando Pastawski[1], Terry Rudolph[1] & Chris Sparrow[1]



	Success	Failure	Erasure
Stabilizers measured	X_1X_2 Z_1Z_2	Z_1Z_2	None







Loss has to be very low…

nature

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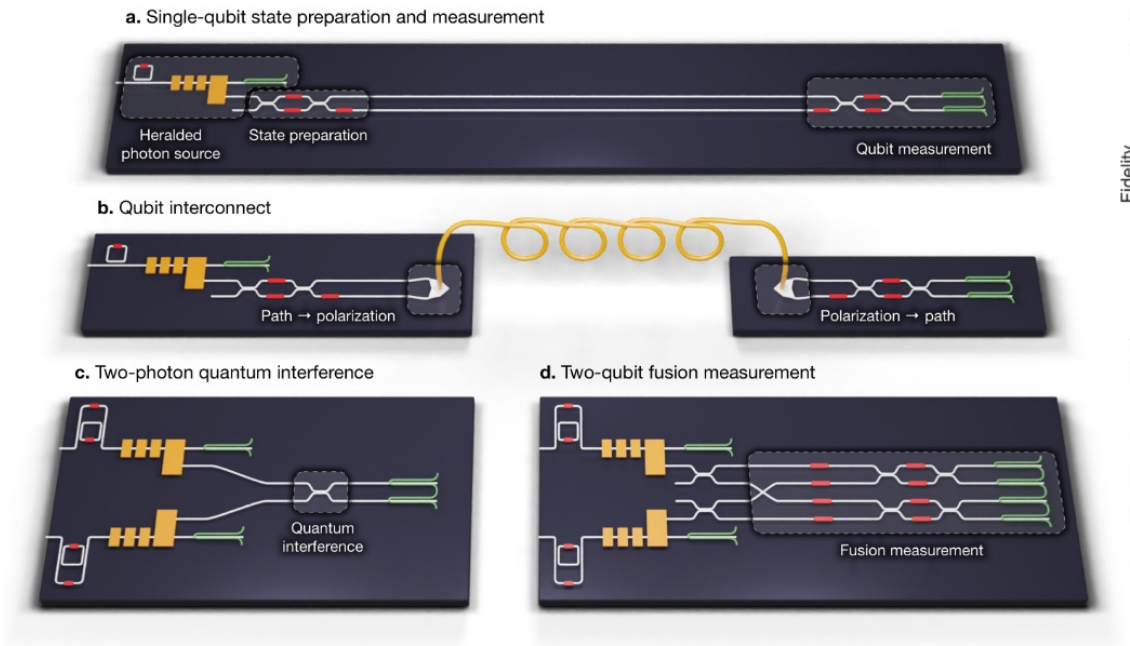
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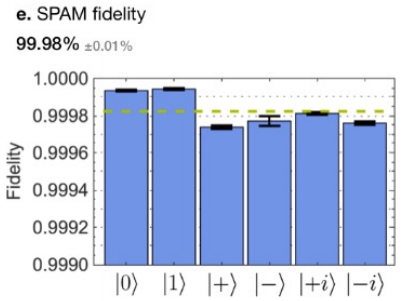
Article | Open access | Published: 26 February 2025

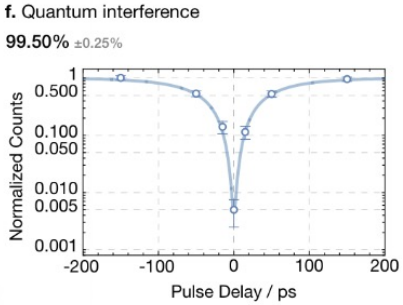
A manufacturable platform for photonic quantum computing

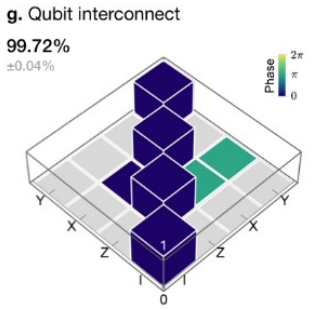
PsiQuantum team

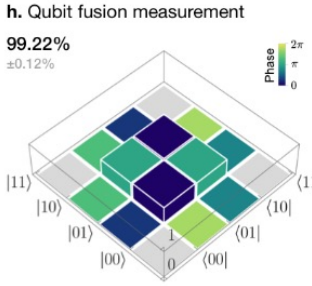
Nature **641**, 876–883 (2025) | Cite this article


a. Single-qubit state preparation and measurement
Heralded photon source
State preparation
Qubit measurement
b. Qubit interconnect
Path → polarization
Polarization → path
c. Two-photon quantum interference
Quantum interference
d. Two-qubit fusion measurement
Fusion measurement


e. SPAM fidelity
99.98% ±0.01%
Fidelity
1.0000
0.9998
0.9996
0.9994
0.9992
0.9990
|0⟩ |1⟩ |+⟩ |−⟩ |+i⟩ |−i⟩


f. Quantum interference
99.50% ±0.25%
Normalized Counts
1
0.500
0.100
0.050
0.010
0.005
0.001
-200 -100 0 100 200
Pulse Delay / ps

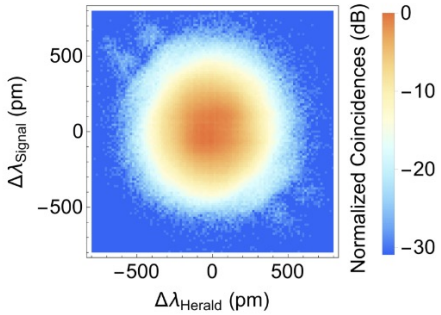

g. Qubit interconnect
99.72%
±0.04%
Phase
2π
π
0
Y
X
Z
I
1
0


h. Qubit fusion measurement
99.22%
±0.12%
Phase
2π
π
0
|11⟩
|10⟩
|01⟩
|00⟩
⟨11|
⟨10|
⟨01|
⟨00|

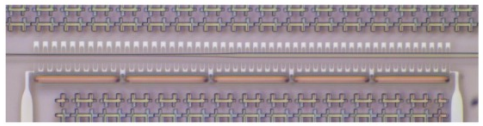
a. Cascaded resonator source



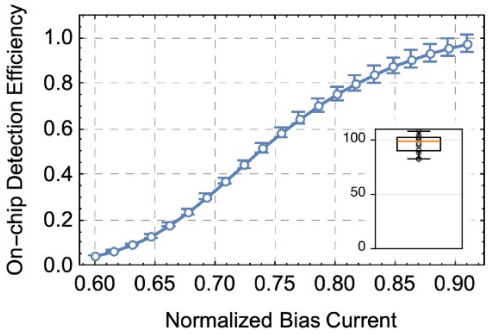
b. Source JSI



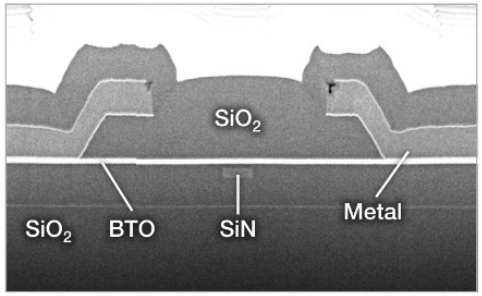
d. Photon number resolving detector



e. PNRD on-chip efficiency



e. BTO phaseshifter cross-section



g. BTO MZI voltage response

